

Curated Research Articles

Generated: 2026-06-01 07:01

- **Molecular Design Strategies for High-Voltage Organic Cathodes** — score: 1.000 This study develops molecular design strategies to enhance the performance of high-voltage organic cathodes for lithium-ion batteries, specifically focusing on phenazine-1,4,6,9-tetrone (PzTO) and dithiin-fused naphthazarin (DNP). By analyzing factors such as Li-O coordination and changes in aromaticity during reduction, the research identifies key features that influence redox potentials and suggests a framework for optimizing organic electrode materials for improved battery efficiency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Operando Imaging of Adsorbate-Charge Cooperative Reconstruction into Terminal Active Co-OH Sites** — score: 1.000 The article discusses an innovative approach to operando imaging, revealing how adsorbate-charge interactions lead to cooperative structural changes in active Co-OH sites. This research provides insights into the dynamics of catalyst surface modifications, enhancing the understanding of their activity in chemical reactions. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Labile Semi-Cage Solvation Structures Enable Efficient Ion Transport in Solid Polymer Electrolytes** — score: 1.000 The article discusses the development of labile semi-cage solvation structures in PEO-based solid polymer electrolytes, achieved through targeted salt and additive engineering. This innovative approach weakens the binding of Li⁺ to the polymer, facilitating rapid ligand exchange and significantly enhancing ionic conductivity, which can advance the performance of lithium metal batteries by overcoming traditional limitations in ion transport. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Restraining Mechanochemical Failure in High-Voltage Ni-Rich Cathode by Coupling Bulk Lattice Pinning With Interphase Self-Reconstruction** — score: 0.900 The article presents a novel strategy that enhances the stability of high-voltage Ni-rich cathodes by combining bulk lattice pinning with a self-reconstructing interphase. This approach mitigates both intra- and intergranular cracking through reduced strain and protective layers, ultimately improving the performance and longevity of lithium-ion batteries, with remarkable capacity retention demonstrated after extensive cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Beyond Electrostatic Interactions: Ring Strain-Driven Weakening of Solvent-Li⁺ Binding Enables Facile FSI Anion Reduction and Inorganic-Rich SEI Formation in Fluorinated Ether Electrolytes** — score: 0.800 This article explores how ring strain in fluorinated ether electrolytes affects the binding of Li⁺ ions, reducing electrostatic interactions. This weakening facilitates the reduction of FSI anions and leads to the formation of a more inorganic-rich solid-electrolyte interphase (SEI), ultimately enhancing the performance of battery electrolytes. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Operando Decoding of Stress Relaxation Behavior in LiFePO₄/Graphite Pouch Cells during Resting States** — score: 0.800 The study investigates the stress relaxation behavior of LiFePO₄/Graphite pouch cells during resting periods using operando techniques. It reveals insights into how structural changes and mechanical stresses evolve over time, which is crucial for improving the performance and longevity of battery systems. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Synergistic Decomposition of Closo-Type Borohydrides Enables Robust Cathode-Electrolyte Interphases in Ah-Class Lithium Cells** — score: 0.800 The study presents Li₂B₁₂H₁₂ as an innovative electrolyte additive that enhances the cathode-electrolyte interphase (CEI) in lithium metal batteries. Through a unique decomposition process producing a robust CEI rich in borides and fluorides, the additive supports high voltage stability and significantly improves the electrochemical performance of various cathodes, achieving an energy density of 392 Wh kg⁻¹ in a 3 Ah-class pouch cell. Journal: *Wiley: Small Methods: Table of Contents*

- **Hypochlorite Redox Chemistry Enables High-Voltage and High-Power Saltwater Batteries** — score: 0.800 This study presents a novel approach to enhance the performance of saltwater batteries (SWBs) by replacing traditional oxygen redox reactions with sodium hypochlorite-based redox chemistry. The new system achieves higher discharge voltages and significantly improved power density while maintaining stability and scalability, making it a promising alternative for large-scale stationary energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Suppressing Dendrites via Lateral Lithium Flux in Li Metal Solid-State Batteries** — score: 0.800 The article discusses a novel approach to mitigate lithium dendrite growth at the lithium-solid-state electrolyte interface, a critical issue for the advancement of lithium metal solid-state batteries. By utilizing lateral lithium flux, the researchers aim to suppress dendrite formation, thereby enhancing battery safety and performance. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Metal electrode potential diverges with ion additions** — score: 0.800 The study reveals that the coordination environment of ions significantly affects the redox potentials of metal ions by altering their electrostatic potential, driven by the hard/soft character of these coordinating ions. This understanding provides a framework for the rational design of electrolytes in electrochemical applications, leading to enhanced performance. Journal: *Nature Chemistry*
- **High-Performance Elastomeric Lithium Metal Anodes Enabled by a Lithiophilic Monolayer-Assembled Nano-Crumpled Micro-Concave Architecture** — score: 0.800 The study introduces a flexible lithium metal anode featuring a unique nano-micro architecture enhanced with lithiophilic functionalities, which significantly mitigates dendrite formation during cycling. This innovative design not only stabilizes the anode's performance, allowing over 2,100 hours of cycling at high current rates but also maintains approximately 90.2% capacity retention in full cells after 1,000 cycles, positioning elastomeric anodes as a viable solution for safer and longer-lasting lithium metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A Focused Review on the Current Collector to Enable High-Performance Soft-Gel Electrode Batteries** — score: 0.800 This review repositions current collectors in batteries from passive roles to dynamic regulators of electrochemical processes, impacting ionic migration and reaction stability. It examines how design factors such as geometry and surface chemistry influence performance, and proposes a framework for developing advanced current collectors to enhance the efficiency of soft-gel electrode batteries, highlighting their potential for integrating multiple functions to improve battery performance and safety. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic Electrochemical and Mechanical Enhancement in Structural Batteries based on a Porous Resin** — score: 0.800 The article discusses the development of a high-performance structural battery composite using a two-step structural battery electrolyte (SBE) made from a porous resin. This innovative approach not only improves lithium-ion transport and cycling stability but also maintains structural rigidity, resulting in a battery that achieves excellent energy density and mechanical strength. The findings suggest a promising direction for future advancements in the design and scalability of structural batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Reversibility-Driven Degradation Mechanisms and Targeted Modulation Strategies for Anode-Free Sodium Metal Batteries** — score: 0.800 This review examines the degradation mechanisms in anode-free sodium metal batteries (AFSMBs), highlighting the irreversible depletion of sodium due to unstable interfacial chemistry. It categorizes sodium loss pathways based on theoretical reversibility and discusses strategies for mitigating capacity fade, focusing on uniform sodium deposition and solid electrolyte interphase regulation while identifying challenges and future research directions in enhancing AFSMB performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic Enhancement to Polysulfide Catalytic Conversion and Lithium Dendrite Resistant by 3D Cross-Linked Fe-N4-GCs for Stable Lithium-Sulfur Batteries of Low-N/P-Ratio** — score: 0.800 The article reports on the development of a novel lithium-sulfur battery using 3D cross-linked iron-nitrogen sites on graphene-like porous carbon as a catalyst and lithium regulator. This design enhances the conversion of lithium polysulfides and mitigates lithium

dendrite formation, resulting in impressive battery performance with high initial capacity, stable cycling over 1000 cycles, and a low N/P ratio. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Stepwise Elemental Screening of High-Entropy Oxides as Zero-Strain and High-Capacity Anode in LIBs** — score: 0.800 This study introduces a method for efficiently screening high-entropy oxides (HEOs) as potential anode materials for lithium-ion batteries (LIBs) by focusing on formation energy and volume change metrics. The breakthrough anode material, $(\text{CoFeMnNiZnMgLi})_3\text{O}_4$, demonstrates remarkable stability with only a 0.38% volume change and negligible capacity decay over 3000 cycles, highlighting its potential for advanced energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **2D Thermal Imaging of Large-Format Batteries Using Distributed Optical Fiber Sensing and Physics-Informed Neural Networks** — score: 0.800 The article discusses a method for 2D thermal imaging of large-format batteries utilizing distributed optical fiber sensing combined with physics-informed neural networks. This innovative approach aims to enhance thermal management and safety in battery systems, thereby contributing to their efficiency and longevity. Journal: *ScienceDirect Publication: Nano Energy*
- **Steam-induced cation exchange in oxygen electrodes for superior protonic solid oxide cells** — score: 0.700 The article presents a steam-engineered $\text{BaCo}_{0.8}\text{Fe}_{0.1}\text{Hf}_{0.05}\text{W}_{0.05}\text{O}_3$ — oxygen electrode that enhances the performance of protonic solid oxide cells through dynamic tungsten-iron cation exchange. Detailed studies using various analytical techniques reveal the mechanism of proton-involved oxygen reduction reactions, while the electrode’s scalable fabrication indicates its manufacturability and potential for widespread use in fuel cells and electrolyzers. Journal: *Joule*
- **Blocking oxidation of -hydrogens enables non-fluorinated solvents to achieve high-potential stability in lithium batteries** — score: 0.700 Recent research published in *Nature Chemistry* demonstrates that inhibiting -oxidation in non-fluorinated carboxylate solvents can significantly enhance high-potential stability in lithium batteries. This advancement allows for the development of batteries exceeding 650 Wh/kg without the need for costly and environmentally damaging fluorinated solvents. Journal: *Nature Chemistry*
- **The Stability of Electrodes with Intermediate Layer-Induced Catalytic Layer-Matrix Layer Interfaces in Chlorine Evolution Reactions** — score: 0.600 The study presents a novel $\text{RuSnOx}/\text{IrTaOx}/\text{Ti}$ mesh electrode designed for chlorine evolution reactions, showcasing high activity and exceptional stability. The interlayer of IrTaOx enhances the oxidation state of Ru to improve reaction efficiency while also preventing dissolution of the active layer and oxidative damage to the titanium substrate, resulting in a durable and efficient catalyst for industrial applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Advanced Battery Technologies and the Role of 3D/4D Printing in Their Evolution** — score: 0.500 The review discusses the transformative impact of 3D and 4D printing on advanced battery technologies, emphasizing their ability to create customizable and adaptive battery architectures that improve performance and sustainability. It explores various battery chemistries, highlighting the role of smart materials in enhancing safety and flexibility while addressing challenges in scalability and material stability. The review also advocates for the integration of recycling practices to enhance environmental benefits and promote the development of next-generation energy storage systems. Journal: *Wiley: Small Methods: Table of Contents*
- **[ASAP] Ketjen Black-Decorated $\text{Li}_2\text{C}_4\text{O}_4$ for LiF-Rich CEI Engineering and Mn Dissolution Suppression in Long-Life $\text{LiMn}_{0.6}\text{Fe}_{0.4}\text{PO}_4$ Cathodes** — score: 0.500 The article discusses the development of Ketjen Black-decorated $\text{Li}_2\text{C}_4\text{O}_4$, which enhances the solid electrolyte interphase (SEI) in lithium manganese iron phosphate ($\text{LiMn}_{0.6}\text{Fe}_{0.4}\text{PO}_4$) cathodes. This innovation aims to improve the cycling stability and lifespan of the batteries by suppressing manganese dissolution and promoting a lithium fluoride-rich composition in the cathode environment. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*

- **Reversible self-assembly of polymeric solid electrolyte enables lithium battery recycling** — score: 0.500 The article discusses a novel supramolecular electrolyte that enhances the performance and recyclability of solid-state batteries. This electrolyte solidifies during battery operation and can revert back to its molecular form after use, addressing significant challenges in battery recycling while maintaining high energy storage capabilities. Journal: *Nature Chemistry*
- **Pyridine-Isomerism-Driven Spin Engineering at a COP-Fe₃O₄ Interface for High-Performance Oxygen Reduction Electrocatalysis** — score: 0.500 The article discusses a novel strategy involving pyridine isomerism to enhance the interaction between covalent organic polymers (COPs) and Fe₃O₄ nanocrystals, leading to an improved spin state of iron. This optimization significantly boosts the electrocatalytic activity, kinetics, and stability of the oxygen reduction reaction (ORR) in alkaline conditions, with the engineered COP-Fe₃O₄ composite showing superior performance in alkaline electrolytes and aqueous Zn-air batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Battery Safety: Mechanisms, Monitoring, and Machine Intelligence** — score: 0.500 The article discusses an integrated approach to battery safety, highlighting the interplay between failure mechanisms, intrinsic material safety, and advanced monitoring systems that utilize machine intelligence. It emphasizes the importance of multiphysics understanding, cloud-edge monitoring, and AI-driven management to create proactive, interpretable safety controls for diverse battery chemistries and usage conditions. The review also proposes future directions for safety standards, autonomous management, and innovative technologies aimed at enhancing battery reliability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Ultrafast ion-exchange synthesis of 3D lithiophilic scaffolds for high-capacity anode-less all-solid-state batteries** — score: 0.400 The article discusses the development of a rapid ion-exchange method to create three-dimensional lithiophilic scaffolds, which significantly enhance the performance of anode-less all-solid-state batteries (ALASSBs). These innovations aim to address challenges in achieving high-capacity and efficient energy storage, thereby improving the viability of ALASSBs for practical use. Journal: *RSC - EES Batteries latest articles*
- **[ASAP] Ti⁴⁺ Substitution at the Mn Site Enhances Na⁺ Diffusion and Cycling Stability of O₃-NaNi_{0.33}Fe_{0.33}Mn_{0.33}O₂ Cathodes for Sodium-Ion Batteries** — score: 0.400 The study investigates the impact of Ti⁴⁺ substitution at the manganese sites in O₃-NaNi_{0.33}Fe_{0.33}Mn_{0.33}O₂ cathodes, revealing that this modification significantly enhances sodium ion diffusion and cycling stability. The findings suggest that tailored cation substitution can improve the performance of sodium-ion batteries, paving the way for more effective energy storage solutions. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Gradient Interphase for synergistic optimization of dendrite-free Zn anode and anti-corrosion halogen cathode to enable long-cycling aqueous Zn-iodine battery** — score: 0.400 The article presents a novel gradient interphase design that enhances the performance of aqueous Zn-iodine batteries by preventing dendrite formation on the zinc anode and improving the corrosion resistance of the halogen cathode. This development aims to enable prolonged cycling stability for these batteries, addressing key challenges in their efficiency and longevity. Journal: *ScienceDirect Publication: Nano Energy*
- **[ASAP] High-Performance Na_{1.9}K_{0.1}VTi(PO₄)₃-GNP with Multielectron Redox for Printed Aqueous Symmetric Sodium-Ion Batteries** — score: 0.300 The article presents the development of a high-performance sodium-ion battery cathode material, Na_{1.9}K_{0.1}VTi(PO₄)₃-GNP, which utilizes multielectron redox reactions to enhance energy storage. This new material shows promise for use in printed aqueous symmetric sodium-ion batteries, potentially improving efficiency and sustainability in energy storage applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Spider Dragline Silk-Inspired 3D Printed Sustainable Zinc Ion Battery with Ultrahigh Stability** — score: 0.300 The study presents a novel zinc-ion battery inspired by spider dragline

silk, utilizing 3D printing techniques to enhance stability and sustainability. It addresses significant challenges such as dendrite formation, which hampers the battery’s performance, aiming to provide an efficient solution for large-scale energy storage with environmental benefits. Journal: *RSC - Energy Environ. Sci. latest articles*

- **Suppressing Polysulfide Shuttling via A Supporting Electrolyte Concentration-Asymmetric (SEC-Asym) Design based on Electrolyte Ionic Strength for Stable Promoted Na–Polysulfide Redox-Flow Batteries** — score: 0.300 The article presents a novel approach to enhancing the stability of Na–polysulfide redox-flow batteries by implementing a concentration-asymmetric electrolyte design that mitigates polysulfide shuttling. The study highlights the importance of electrolyte ionic strength in improving battery performance and longevity, addressing a key challenge in energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Synthetic Pathways in Mg-Al-based LDH Systems to a Versatile Library of Single-layer Porous Derivatives With Structural Transformations and Tunable Doping** — score: 0.200 The article presents a novel synthetic framework for transforming Mg-Al-based layered double hydroxides (LDHs) into a diverse range of single-layer porous materials, including oxides, fluorides, and oxysulfides. This method allows for tunable doping, which enhances the multifunctional properties of the materials, demonstrating the ability to systematically link matrix composition with functionality such as luminescence and magnetic properties. Journal: *Wiley: Small Methods: Table of Contents*
- **Cutting-Edge Advances in Metal Phosphides: Rational Synthesis to Dynamic Reconstruction for Electrochemical Water Splitting** — score: 0.200 This review examines the advancements in transition metal phosphides (TMPs) for electrochemical water splitting, emphasizing efficient microwave-assisted synthesis and the modifications that enhance their catalytic performance. It discusses the mechanisms behind their activity, the importance of dynamic reconstruction during operation, and highlights their potential for scalable hydrogen production. The article also addresses current challenges and future perspectives in leveraging TMPs within green energy systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Enhanced stability of extended viologen anolyte for aqueous organic redox flow batteries via an electron-withdrawing imidazole group** — score: 0.200 This study presents the development of an enhanced anolyte for aqueous organic redox flow batteries by incorporating an electron-withdrawing imidazole group into extended viologen structures. The modification significantly improves the stability of the viologen derivatives, promising greater efficiency and longevity for energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] A Small-Molecule Naphthoquinone-Based Organic Cathode Material for High-Performance Lithium-Ion Batteries** — score: 0.200 The article discusses the development of a novel small-molecule naphthoquinone-based organic cathode material designed for lithium-ion batteries. This new material demonstrates enhanced performance characteristics, potentially improving the efficiency and longevity of battery systems. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] An Approach to Modifying Photoelectrodes with Nickel Foam for Enhanced Performance in Solar Redox Flow Batteries** — score: 0.200 The article proposes a novel method for enhancing the performance of solar redox flow batteries by modifying photoelectrodes with nickel foam. This modification aims to improve efficiency and energy output, showcasing potential advancements in renewable energy storage technologies. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Flame-Retardant Self-Healing Flexible Poly(siloxane amine) Quasi-Solid-State Electrolyte for High-Energy-Density Lithium-Metal Batteries** — score: 0.200 The article discusses the development of a flame-retardant self-healing quasi-solid-state electrolyte based on poly(siloxane amine), designed to enhance the safety and performance of high-energy-density lithium-metal batteries. This innovative electrolyte aims to address the challenges of conventional

electrolytes by offering improved thermal stability and the ability to self-repair, potentially leading to longer-lasting and safer battery systems. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*

- **[ASAP] Leveraging Water/Ethanol Solvent Ratios to Tailor the Properties of Sodium Carboxymethylcellulose and Ionic Liquid— Solid Polymer Electrolytes for Solid-State Batteries** — score: 0.200 The article explores the use of varying water/ethanol solvent ratios to modify the properties of sodium carboxymethylcellulose and ionic liquid-based solid polymer electrolytes, aiming to enhance the performance of solid-state batteries. This approach may lead to improved conductivity and stability in battery applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Janus Separator with a Graphene Mesosponge Empowers High-Rate Zn-Ion Hybrid Capacitors** — score: 0.200 The article discusses a newly developed Janus separator integrated with a graphene mesosponge that enhances the performance of zinc-ion hybrid capacitors. This innovation enables high-rate operation, potentially improving energy storage and efficiency in applications relying on this technology. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Self-Adaptive Stability in High-Capacity SiO_x Anodes Enabled by a Dynamic Bilinear Interpenetrating Binder** — score: 0.200 The article discusses the development of high-capacity silicon oxide (SiO_x) anodes for lithium-ion batteries, highlighting how a dynamic bilinear interpenetrating binder enhances their self-adaptive stability. This innovative approach aims to improve the performance and reliability of energy storage systems. Journal: *ScienceDirect Publication: Nano Energy*
- **High-Performance LiFePO₄ Regeneration through Na-Induced Lattice Activation and Conductivity Rewiring** — score: 0.200 This article discusses a novel approach to regenerate spent LiFePO₄ (LFP) batteries using sodium to activate the lattice structure and enhance conductivity, offering a more effective method than traditional lithium-replenishment techniques, which only restore the original performance. This advancement is significant for improving the sustainability of lithium-ion battery technologies. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Developing a Vacancy-Regulated Integrated Cathode for Catalysis–Mass Transfer Coupling Optimization and Interfacial Kinetics Enhancement in Air Batteries** — score: 0.200 The article discusses the development of an advanced integrated cathode designed to optimize both catalysis and mass transfer in air batteries. It focuses on enhancing interfacial kinetics through a vacancy-regulated approach, aiming to improve the overall efficiency and performance of these energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Innovative Chromium–Bromide Flow Batteries: Enabling High–Capacity Energy Storage with Superior Electrochemical Efficiency** — score: 0.200 The article discusses the development of chromium-bromide flow batteries, which offer enhanced energy storage capacity and improved electrochemical efficiency. The authors highlight the innovative design and potential applications of these batteries in advancing energy storage technologies, with findings relevant for future energy management solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Fe Single Atom Enabled Ni Dual Sites Unlock the Oxide Pathway for Efficient Oxygen Evolution** — score: 0.200 This study demonstrates how integrating a Fe atom into Ni dual centers (Ni₂-Fe) enhances the oxygen evolution reaction (OER) by facilitating the transition to an oxide pathway mechanism. The modulation of geometric and electronic properties in the catalyst significantly lowers reaction barriers, enabling efficient OER performance with record-high mass activity at minimal Ni loading, thus presenting a promising strategy for improving electrocatalysts. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Cobalt-free and high-rate stable 5V lithium nickel manganese oxide spinel cathodes enabled via surface oxygen vacancies** — score: 0.200 The article presents advancements in cobalt-free lithium nickel manganese oxide spinel cathodes that demonstrate high-rate stability at 5V due to the introduction of surface oxygen vacancies. Researchers highlight how these modifications enhance the

performance and longevity of cathodes, offering a promising alternative for energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **[ASAP] Chiral Covalent Organic Framework for Highly Efficient Spin Polarization** — score: 0.000 The article presents the development of a chiral covalent organic framework that demonstrates exceptional spin polarization efficiency. This advancement could have significant implications for spintronic applications, enhancing the functionality of devices that utilize electron spin. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Cation-Limited Hydroxide Anion Diffusion Drives Asymmetric Hydrogen Kinetics on Transition-Metal Decorated Platinum Surface** — score: 0.000 The article investigates the influence of cation-limited hydroxide ion diffusion on hydrogen kinetics on platinum surfaces decorated with transition metals. It finds that this diffusion process leads to asymmetric reaction behaviors, which could have implications for optimizing catalytic processes in electrochemical applications. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Theory-Guided Synthesis of Stable Low-Nuclearity Cluster Catalysts via Atom-Stabilizer Locking** — score: 0.000 The article discusses a novel approach for creating stable low-nuclearity cluster catalysts through a method called atom-stabilizer locking, which is guided by theoretical insights. This synthesis technique aims to enhance the efficiency and stability of catalysts, offering potential advancements in catalytic processes. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Single-Atom Synergy of Luminescence and Catalytic Stability in Silver-Encapsulated Gold Nanoclusters** — score: 0.000 This article explores the unique properties of silver-encapsulated gold nanoclusters, focusing on their enhanced luminescence and exceptional catalytic stability. The findings highlight how the incorporation of silver at the atomic level synergistically improves the performance of these nanoclusters, offering insights for future applications in nanotechnology and materials science. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **How I eavesdrop on frog conversations** — score: 0.000 Billie Goolsby, drawing on her personal experience with hearing loss, has created a robotic tadpole designed to listen in on and analyze the communication patterns among frogs, shedding light on the hidden aspects of their social interactions. This innovative approach not only enhances our understanding of amphibian behavior but also showcases the potential of technology in biological research. Journal: *Nature*